

ZeroDrift MK2: belt upgrade and full parts reference

Team ZeroDrift · SIH 2026 PS26169 · **OPTIONAL, AFTER the direct-drive rig works.** For today's shopping use the short list: [ZeroDrift_MK2_Amar_Shopping_List.pdf](#). This document is the long reference: every part, the analysis of the reference turrets, and the GT2 belt on tilt (section C and page G).

Hi, we are building a 2-motor pan-tilt camera turret with NEMA17 stepper motors (5 mm shaft) and A4988 drivers. We need the parts below. Items marked **MUST** are needed for the basic build. **BELT** items are for a GT2 timing-belt gear-down on the tilt motor.

A · Must buy: basic build without these the rig cannot be assembled

#	ASK FOR	EXACT SPEC / CHECK AT THE COUNTER	QTY	≈ ₹
☐ 1	5 mm rigid flange coupling also called "5 mm shaft flange coupler / wheel hub for 5 mm shaft"	Bore 5 mm (a NEMA17 shaft must slide in snugly). Rigid metal, not the flexible/spring type. Has a grub screw. Flange has 4 holes (M3).	2	160–300
☐ 2	M3 brass standoff, 40 mm or 20 mm ×4 (male–female, they screw together)	Brass (gold), M3 thread. Carries the camera head. Not plastic/nylon.	2	40–80
☐ 3	M3 standoff, 15 mm (brass or nylon)	Raises the tilt L-bracket above the pan encoder	4	30–60
☐ 4	M3 screws + nuts + washers pack	Lengths 8, 10, 16, 20 mm , about 10 of each; 30 nuts; 20 washers	1 set	60–120
☐ 5	Resistor 100 Ω, ¼ W brown-black-brown-gold	Decoy LED. Buy 5 (they cost ₹1 each).	5	5–10
☐ 6	Resistor 220 Ω, ¼ W red-red-brown-gold	Beacon LED. Buy 5.	5	5–10
☐ 7	Red LED, high-brightness, 10 mm (5 mm OK)	A second red LED for the decoy . Through the camera's red filter a white LED looks dim, so a steady red decoy is the fair test. Buy 2.	2	10–20
☐ 8	Electrolytic capacitor 100 µF, 25 V (or 35 V)	One across each A4988 motor supply. Buy 3 (1 spare). Skip if you already have 2.	3	15–30
☐ 9	Clear acrylic sheet 3 mm, A4 size	For the pan disc, tilt disc, encoder arms and head plate. Your white sheet is fine for the base ; the small parts must be stiff.	1	120–200
☐ 10	Red transparent acrylic 2 mm (small piece) or red cellophane / red gel sheet	Filter over the camera lens so bright rooms don't blind it. Hold it over your phone camera: the room should look dim red.	1	20–80
☐ 11	Allen key set (1.5–4 mm)	For the grub screws on the couplings and pulleys	1	60–120

B · Power and charging your B3 charger needs a balance plug the 18650 holder does not have

#	ASK FOR	EXACT SPEC / CHECK AT THE COUNTER	QTY	≈ ₹
☐ 12	12 V 2 A DC adapter (SMPS, wall plug) MUST	Output 12 V, 2 A , 5.5 × 2.1 mm barrel plug. Powers the motors on the bench and at the demo table, so the 18650s are only a backup.	1	200–350
☐ 13	DC female barrel jack to screw terminal	5.5 × 2.1 mm, matches the adapter. Screw terminals marked + and –.	2	30–60
☐ 14	TP4056 18650 charging module with protection (USB-C or micro-USB)	Charges one 18650 at a time, out of the holder. Needed because the B3 charger only charges through a balance plug.	1	30–60
☐ 15	Hook-up wire 22 AWG , red + black	1 m each, for the 12 V motor supply (jumper wires are too thin for this)	2 m	30–60
☐ 16	USB cable for the 2nd Nano	Check the Nano's port first: Mini-USB or USB-C. Skip if you have one.	1	60–120

C · Belt gear-down on tilt **BELT** same idea as both turrets in your zip

Gives tilt **3× finer steps** and **3× more holding torque**. The small pulley goes on the tilt motor and the big pulley on a separate 8 mm axle that carries the head. Buy these now so you don't need a second trip. Build the direct-drive rig first and add the belt once it works (page 4).

#	ASK FOR	EXACT SPEC / CHECK AT THE COUNTER	QTY	≈ ₹
☐ 17	GT2 timing pulley, 20 teeth, 5 mm bore	For 6 mm wide belt. Bore 5 mm (fits the NEMA17 shaft). Aluminium with 2 grub screws.	1	80–150
☐ 18	GT2 timing pulley, 60 teeth, 8 mm bore	For 6 mm belt. Bore 8 mm (for the 8 mm axle). If they only have 60T with 5 mm bore: take it, and buy a 5 mm rod and KFL005 bearings instead of 8 mm ones (same shape).	1	200–400
☐ 19	GT2 closed-loop timing belt, 6 mm wide the "rubber" (a ring, no joint)	Length 232 mm (first choice). Also OK: 280 mm or 200 mm. If cheap, buy two different lengths. Must be GT2 (2 mm tooth pitch), same as the pulleys.	1–2	80–150 each
☐ 20	KFL08 flange bearing (2-bolt, 8 mm bore)	Zinc flange with a bearing inside. 8 mm bore, spins smoothly with no grinding. Bolts onto a flat plate.	2	120–250
☐ 21	8 mm steel rod, 100 mm smooth rod / linear shaft; or an M8 × 100 bolt with a smooth shank	The tilt axle. Straight, smooth, fits the KFL08 and the 60T pulley.	1	60–150
☐ 22	8 mm rigid flange coupling	Bore 8 mm . Bolts the head plate to the tilt axle.	1	100–180
☐ 23	8 mm shaft collar (lock collar with grub screw)	Stops the axle sliding sideways	2	40–80
☐ 24	M4 × 20 screws + nuts	For the KFL08 flanges and the motor-to-plate mounting	8	20–40

☐	25 Aluminium flat plate 3 mm, about 70 × 160 mm (or 5 mm acrylic)	The upright that holds the tilt motor (bottom) and the axle bearings (top). A hardware shop can cut it.	1	60–150
☐	26 Small aluminium angle / L-bracket ($\approx 30 \times 30$ mm)	Bolts the upright to the pan platform	2	20–60

D · Holders and boxes you said you have no boxes

#	ASK FOR	EXACT SPEC / CHECK AT THE COUNTER	QTY	≈ ₹
☐	27 ABS plastic project box / enclosure , $\approx 100 \times 60 \times 25$ mm	One for the beacon (Nano + LED + switch), one for the decoy (3×AA holder + LED + switch). Any small plastic box with a lid works; a tiffin box is fine.	2	80–150 each
☐	28 Laser module holder / clip or a 6–12 mm cable P-clip	Holds the laser on the head plate so it points exactly where the camera looks. Hot glue works too.	1	20–60
☐	29 Double-sided foam tape (thin, strong)	Mounts the breadboard and camera board. Skip if you have some (the red roll in your photo is fine).	—	—

No box is needed for the **head**. Both turrets in your zip use an open **plate**, with the camera board and laser screwed straight onto it. It's lighter and easier to balance. Cut it from the 3 mm acrylic (#9).

E · Small consumables

#	ASK FOR	SPEC	QTY	≈ ₹
☐	30 Superglue (Fevikwik) or 5-minute epoxy (Araldite)	Glues the magnets on the axis centres	1	20–120
☐	31 Zip ties , small (100 mm)	Cable service loops	1 pk	30
☐	32 Heat-shrink tube assortment (or insulation tape)	Covers the soldered LED and resistor legs	1	30–60
☐	33 Solder wire (if you have none)	For the TCA9548A and AS5600 header pins	1	40–80
☐	34 USB-C to USB-A hub only if the laptop has USB-C ports only	The Nano and the webcam must be plugged in at the same time	1	300–800

Estimated total, A + B + D + E (without the hub): ≈ ₹1,100–2,000 · with the belt parts (C): add ≈ ₹1,000–1,800

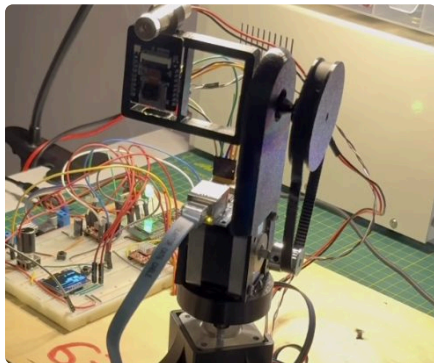
You also need a drill (about 10 holes: 3 mm, 4 mm, 5–6 mm, 8–10 mm). If nobody at home has one, ask Amar or any hardware shop to drill the marked holes. Mark them first with the motor and bearings as templates.

F · Already have: don't buy again

- ✓ 2× NEMA17 stepper motors
- ✓ 2× A4988 drivers (with heatsinks)
- ✓ 2× NEMA17 L-brackets
- ✓ TCA9548A I²C multiplexer
- ✓ 2× AS5600 encoders + diametric magnets
- ✓ 2× Arduino Nano (rig + beacon)
- ✓ KY-008 laser
- ✓ Red LED + white LED
- ✓ 3×AA battery holder
- ✓ Ping-pong ball (beacon diffuser)
- ✓ 3× 18650 cells + 3-cell holder
- ✓ iMAX B3 Pro balance charger
- ✓ Toggle switches
- ✓ Short brass standoffs, M3 screws/nuts
- ✓ Breadboard + jumper wires
- ✓ Multimeter, soldering iron, glue gun
- ✓ White sheet (base plate)
- ✓ USB webcam (arriving)

Charging the 18650s: the B3 charges only through a white 4-pin balance plug, which your holder doesn't have. Take the cells out and charge them **one at a time** on the TP4056 (#14). Before putting them back, measure each: **4.1–4.2 V, all within 0.1 V**. Or run the motors from the 12 V adapter (#12) and skip the battery.

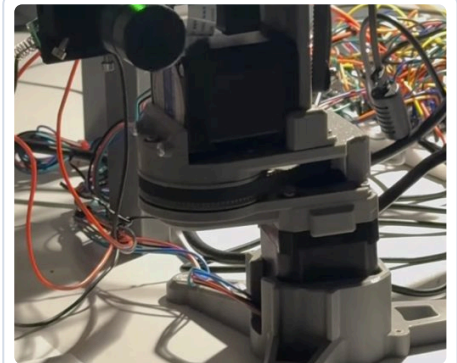
What the turrets in your zip do (analysed)



Turret 1 (AI Laser Turret v1.3, black).
Pan: NEMA17 standing in a base, **direct drive**. Tilt: motor low on the arm, **GT2 belt** from a small pulley up to a big pulley ($\approx 3\text{--}4\text{:}1$). Camera and laser on one open frame.



Turret 2 (grey, green laser), tilt. Same as turret 1: small pulley on the tilt motor, big pulley on the head axle, GT2 belt, $\approx 4\text{--}5\text{:}1$. Camera board and laser on one flat plate.

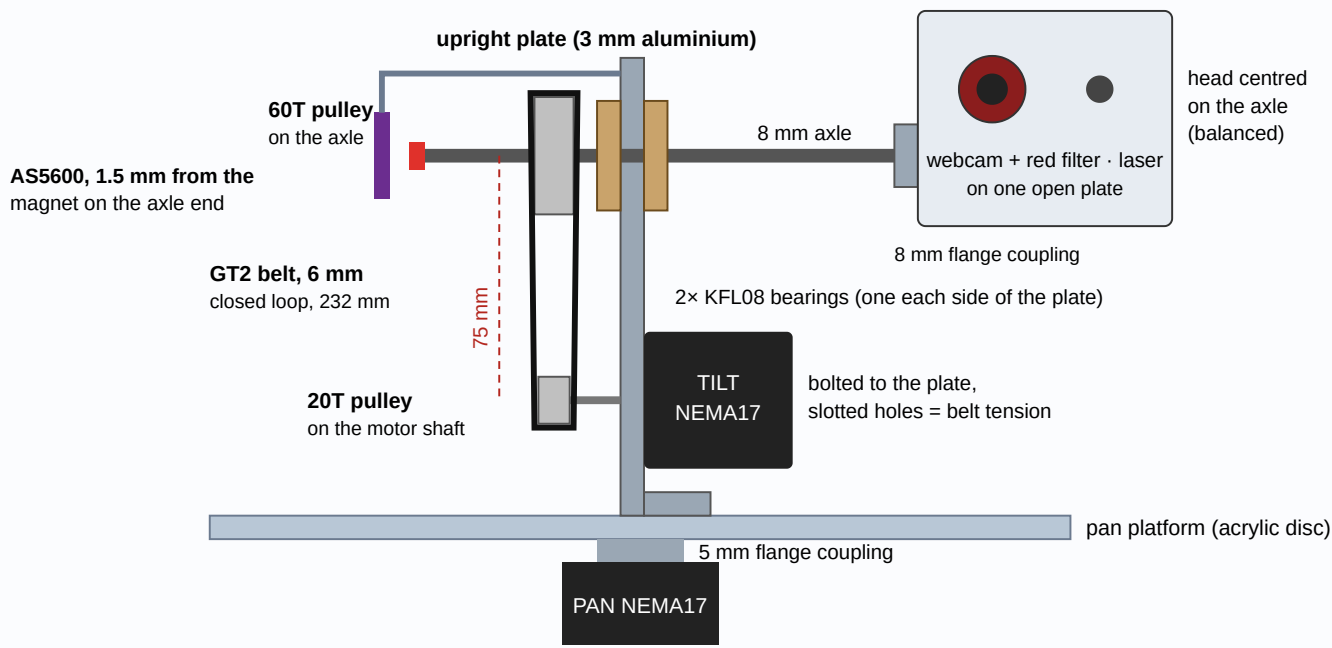


Turret 2, pan. The black band under the upper motor is a **belt around a large turntable pulley**: pan is geared down too. It needs a printed turntable and a large bearing, so we skip it for Monday.

Verdict: both are our MK2 with belts added. We copy three things: **(1) the tilt belt 3:1, (2) the camera and laser on one open plate, (3) the head's weight centred on the tilt axle** (balanced, so the motor doesn't strain). Pan stays **direct drive**: our pan already has an encoder that measures it, and a pan belt needs a big printed turntable we can't make by Monday.

G · Changes to our MK2 model

Belted tilt, seen from the front (the beacon's view). Pan below is unchanged.



What changes, and what stays

PART	BEFORE (DIRECT DRIVE)	NOW (BELTED TILT)
Tilt drive	Head on the tilt motor shaft	Motor low on an upright plate. 20T → 60T GT2 belt , 3:1. Head on a separate 8 mm axle in 2× KFL08 bearings
Tilt step size	0.11° per step	0.0375° per step (3× finer), 3× the holding torque
Tilt encoder	Magnet on a disc on the motor shaft	Magnet glued on the end of the 8 mm axle . It reads the head itself, so any belt slip is caught
Head	Box on a brass standoff	Open plate on an 8 mm flange coupling, weight centred on the axle
Pan	Unchanged: direct drive on a 5 mm flange coupling, AS5600 under the platform	
Firmware	<code>TILT_GEAR_RATIO 1.0</code>	Set <code>TILT_GEAR_RATIO = 3.0</code> (keep <code>PAN_GEAR_RATIO = 1.0</code>) and reflash. Travel limits scale automatically (tilt stays ±18°).
Website	MK2 DRIVE: Direct drive	Already added: MK2 DRIVE → " Pan direct · tilt belt 20T → 60T (3:1) "

Belt length → distance between motor shaft and axle (20T → 60T)

BELT (CLOSED LOOP)	200 MM	232 MM	280 MM	300 MM
Shaft-to-axle centre distance	59 mm	75 mm	99 mm	109 mm

Drill the motor holes as **slots** (2–3 mm long, pointing towards the axle) so you can slide the motor down to tension the belt. Press the belt in the middle: it should give about 2–3 mm. Too loose skips teeth; too tight makes the motor buzz.

Plan for Monday: (1) build the direct-drive rig from the build guide and get it tracking. (2) Then swap in the belted tilt: about 2 hours with the parts from section C. The software already supports both, so the demo is safe even if the belt isn't finished.